

Hilton Stambach 7

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1 Yoneda Lemma Review

Let $F : \mathcal{C} \rightarrow \mathcal{S}$ be a functor and $\tau : \mathcal{C}(A, -) \rightarrow F$ a natural transformation. Then the map $\tau \mapsto \tau_A(1_A)$ is a one-to-one correspondence between $[\mathcal{C}(A, -), F] \rightarrow F(A)$.

Proof: Suppose $\tau : \mathcal{C}(A, -) \rightarrow F$ is a natural transformation and $f : A \rightarrow B$. We have the (c) diagram:

$$\begin{array}{ccc} \mathcal{C}(A, A) & \xrightarrow{\tau_A} & F(A) \\ \downarrow \mathcal{C}(A, f) & & \downarrow Ff \\ \mathcal{C}(A, B) & \xrightarrow{\tau_B} & F(B) \end{array}$$

so $Ff\tau_A(1_A) = \tau_B\mathcal{C}(A, f)(1_A) = \tau_B(f)$, so τ is determined by $\tau_A(1_A)$. Now suppose we have $y \in F(A)$. Define $\tau_B(f) = (Ff)(y)$. We have $\tau_A(1_A) = 1_{FA}y = y$. It remains to show that τ is natural. Indeed for $f : B_1 \rightarrow B_2$, we have the cd

$$\begin{array}{ccc} \mathcal{C}(A, B_1) & \xrightarrow{\tau_{B_1}} & F(B_1) \\ \downarrow \mathcal{C}(A, f) & & \downarrow Ff \\ \mathcal{C}(A, B_2) & \xrightarrow{\tau_{B_2}} & F(B_2) \end{array}$$

since for $g : A \rightarrow B_1$, we have $Ff\tau_{B_1}(g) = (Ff)(Fg)(y) = F(fg)(y) = \tau_{B_2}(fg) = \tau_{B_2}\mathcal{C}(A, f)(g)$. As a corollary, we can speak of the set of natural transformations from $\mathcal{C}(A, -)$ to F . Now take $F = \mathcal{C}(A', -)$, so $[\mathcal{C}(A, -), \mathcal{C}(A', -)] \leftrightarrow \mathcal{C}(A', A)$.

2 Adjoint Functors

Suppose $F \dashv G$. Then F determines G up to natural equivalence. Similarly, G determines F up to natural equivalence.

Proof: Suppose $\eta : F \dashv G$, $\eta' : F \dashv G'$. We have the natural equivalence $\mathcal{C}(-, GY) \xrightarrow{\eta^{-1}} \mathcal{D}(F-, Y) \xrightarrow{\eta'} \mathcal{C}(-, G'Y)$, so by the Yoneda lemma, we have an isomorphism $\theta_Y : GY \rightarrow G'Y$. $\eta'\eta^{-1}$ is natural in Y , so θ is a natural equivalence.

3 Direct Limits

Let I, J be directed and for all $j \in J$, associate directed $I_j \leq I$ such that $I = \bigcup_{j \in J} I_j$. Let $\{\phi_{i',i}, M_i\}$ be a direct system. Let $\phi_i^j : M_i \rightarrow M^j$, $\phi^j : M^j \rightarrow M$, where M is the direct limit wrt the system $\{\phi_{i',i}, M_i\}$. Let M^J denote the direct limit of the M^j wrt $\phi^{j',j}$. Then $\phi : M^J \rightarrow M$ is an isom.

Proof: Define $\phi^{j',j} : M^j \rightarrow M^{j'}$ as follows. First, for all $i \in I_j$, we have $\phi_i^{j',j} : M_i \rightarrow M^{j'}$ by picking $i_1 \geq i$ in $I_{j'}$, and we define $\phi_i^{j',j}(x_i) = \phi_{i_1}^{j'} \phi_{i_1,i}$. I claim that $\phi_i^{j',j}$ is well-defined. Suppose I pick i_2 analogous to i_1 . Pick $i_3 \geq i_1, i_2$ in $I_{j'}$. Then

$$\phi_{i_1}^{j'} \phi_{i_1,i} = \phi_{i_3}^{j'} \phi_{i_3,i_1} \phi_{i_1,i} = \phi_{i_3}^{j'} \phi_{i_3,i} = \phi_{i_2}^{j'} \phi_{i_2,i},$$

as desired. I claim that $\{\phi_i^{j',j}, \phi_{i',i}\}$ are compatible. Indeed,

$$\phi_{i'}^{j',j} \phi_{i',i} = \phi_{i_1}^{j'} \phi_{i_1,i'} \phi_{i',i} = \phi_{i_1}^{j'} \phi_{i_1,i} = \phi_i^{j',j},$$

as desired. Thus, we have $\phi^{j',j} : M^j \rightarrow M^{j'}$ that is well-defined.

Define $\phi^j : M^j \rightarrow M$ by combining the maps $\phi_i : M_i \rightarrow M$ for all $i \in I_j$. I claim that $\{\phi^j, \phi^{j',j}\}$ are compatible. Let $x^j \in M^j$, so $x = \phi_i(x_i)$ for some $x_i \in I_j$. Pick $i_1 \geq i$ in $I_{j'}$, so

$$\phi^{j'} \phi^{j',j}(x^j) = \phi^{j'} \phi_{i_1}^{j'} \phi_{i_1,i} x_i = \phi_{i_1} \phi_{i_1,i} x_i = \phi_i x_i = \phi^j(x^j),$$

as desired. Thus, we have a map $\phi : M^J \rightarrow M$.

I claim that ϕ is surjective. Let $x = \phi_i(x_i)$ in M . Pick j such that $i \in I_j$. Then $\phi(\phi_i^j x_i) = \phi_i x_i = x$. Suppose $\phi x = 0$. Pick $x = \phi^j x^j$ and $x^j = \phi_i x_i$ for some $i \in I_j$. Then $\phi x = \phi \phi^j \phi_i x_i = \phi_i x_i = 0$. Pick $i_1 \geq i$ such that $\phi_{i_1,i} x_i = 0$. Pick j_1 such that $i_1 \in I_{j_1}$. Pick $j_2 \geq j, j_1$. Pick $i_2 \in I_{j_2}$ so that $i_2 \geq i_1, i$. We have

$$\begin{aligned} x = \phi^j x^j &= \phi^{j_2} \phi^{j_2,j} x^j = \phi^{j_2} \phi^{j_2,j} \phi_i x_i \\ &= \phi^{j_2} \phi_{i_2}^{j_2,j} \phi_{i_2,i} x_i = \phi_{i_2}^{j_2} \phi_{i_2,i} x_i \\ &= \phi_{i_2}^{j_2} \phi_{i_2,i_1} \phi_{i_1,i} x_i = 0, \end{aligned}$$

so ϕ is injective.

4 Tensor Product

Let A be an abelian group. Then $A \otimes \mathbb{Z}_n = A_n$.

Proof: Define $\tilde{\beta}_0 : A \times \mathbb{Z} \rightarrow A_n$ by $\tilde{\beta}_0(a, k) = \overline{ka}$. For fixed a , we have $\tilde{\beta}_0(a, n) = \overline{na} = 0$, so we have the induced $\beta_0 : A \times \mathbb{Z}_n \rightarrow A_n$. β_0 is surjective. Suppose $\beta : A \times \mathbb{Z}_n \rightarrow B$ is bilinear.

Define $\tilde{f} : A \rightarrow B$ by $\tilde{f}(a) = \beta(a, \bar{1})$. Then $\tilde{f}(na) = \beta(na, \bar{1}) = \beta(a, \bar{n}) = \beta(a, 0) = 0$, so we have the induced $f : A_n \rightarrow B$. I claim that $f\beta_0 = \beta$. Indeed,

$$f\beta_0(a, \bar{k}) = f(\overline{ka}) = \beta(\overline{ka}, \bar{1}) = \beta(a, \bar{k}).$$

claim: let $A' \xrightarrow{\mu} A \xrightarrow{\varepsilon} A'' \rightarrow 0$ be exact. then $M \otimes A' \rightarrow M \otimes A \rightarrow M \otimes A'' \rightarrow 0$ is exact.

proof: suppose $x \otimes a'' \in M \otimes A''$ (these generate $M \otimes A''$). pick a such that $\varepsilon a = a''$, so $(1 \otimes \varepsilon)(x \otimes a) = x \otimes a''$, so $1 \otimes \varepsilon$ is surjective.

$$\begin{array}{ccccccc} M \otimes N' & \longrightarrow & M \otimes N & \longrightarrow & M \otimes N'' & \longrightarrow & 0 \\ & \searrow 0 & \downarrow f & & \nearrow & & \\ & & Q & & & & \end{array}$$

$$\begin{array}{ccccccc} M \times N' & \longrightarrow & M \times N & \longrightarrow & M \times N'' & \longrightarrow & 0 \\ & \searrow 0 & \downarrow \varphi & & \nearrow & & \\ & & Q & & & & \end{array}$$

sps $M'_i \rightarrow M_i \rightarrow M''_i$ is exact for all $i \in I$. Then $\bigoplus M'_i \rightarrow \bigoplus M_i \rightarrow \bigoplus M''_i$ is exact.